Soap versus COVID-19

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ABSTRACT

Soap vs COVID-19. This short cinematic 3D visualization illustrates the power of soap and water in the destruction and removal of the COVID-19 virus from our hands.

The message in the film is intentionally direct and emotive. Using contemporary digital cinematography combined with scientifically accurate 3D data visualization of the full virus structure, this fully computer-generated piece explores the dynamic interactions occurring at the molecular level when we wash our hands with soap and water. Underpinned by a provoking soundtrack with the terrain of the human hand, the aesthetic of this work parallels a barren battlefield, positioning this piece of public information beyond the didactic. The work also draws influence from contemporary sci-fi film making and video game full motion videos (FMVs).

This unique 3D visualization combines the latest scientific 3D data on the structure and composition of the SARS-CoV-2 virion (virus particle in the film) with industry-leading computer graphics pipelines. The molecular structures of the viral proteins including the spike, membrane, and envelope proteins, as well as the nucleoprotein were sourced from the protein data bank (https://www.rcbs.org) and applied using MolecularMaya (Clarafi, https://clarafi.com).

The membrane lipids on the surface of the membrane were created using the online membrane builder - CHARMM-GUI (https://charmm-gui.org), while soap molecules were modeled by a 3D computer artist. Complete virions were created by procedurally populating a spherical base geometry with various proteins or lipids using Autodesk Maya's MASH tool, while the core consisted of nucleoproteins populated along a dynamic curve.

Ultimately, this film is designed to be a public information release that reminds the audience of the importance of handwashing with soap and water during the COVID-19 global pandemic.

ACM Reference Format:
John McGhee, Andrew Lilija, Campbell Strong, Benjamin Bailey, and Pall Thordarson. 2020. Soap versus COVID-19. In *Proceedings of SA '20 Computer Animation Festival.* ACM, New York, NY, USA, 1 page. https://doi.org/10.1145/3414687.3434105

SA '20 Computer Animation Festival, December 04-13, 2020, Virtual Event, Republic of Korea

ACM ISBN 978-1-4503-8109-3/20/11.
https://doi.org/10.1145/3414687.3434105